Prediction of Winning Probability and Ranking Generation in Horse Racing Using Machine Learning on Pre-Race Data

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Abstract*—*Resumen preliminar.
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This section reports the exploratory analysis performed before any model was trained. It is organised around one question: which information was genuinely available when the starting gate opened, and which was not. Feature admissibility, imputation and partitioning all follow from that question. The analysis is reproducible from with a fixed seed; every number below is regenerated by that script.

A. Data Dictionary

The dataset consists of two relational files published by Daley [?], derived from the public race records of the Hong Kong Jockey Club (HKJC). holds one row per race (6 349 rows, 37 columns) and one row per horse–race pair (79 447 rows, 37 columns); the two join on . The window spans 2 June 1997 to 28 August 2005 and covers 4 405 horses, 186 jockeys and 176 trainers at the Sha Tin and Happy Valley courses. One race in has no corresponding runners and is dropped, leaving 6 348 usable races.

Table ?? summarises both files. Each field carries an *availability* label — PRE for information published before the race, MKT for market prices, POST for anything that only exists once the race is decided. That label, not predictive power, determines admissibility. The full dictionary with dtypes, cardinality, ranges and null counts per column is generated into .

B. Data Leakage Audit

Predicting a race from a table that also contains the outcome is trivial and worthless, so the admissibility rule was fixed before any feature was written: *a variable may enter the model only if its value was determined and published before the gates opened for that race.* Applying it column by column across both files discards 48 fields (Table ??), leaving 13 raw pre-race attributes on the runner table and 11 on the race table as the basis for feature engineering.

The target is a deterministic function of (= **1**[= 1]), so keeping would amount to handing the model its label. The sectional blocks describe the race as it unfolds. The payout fields deserve separate mention as the most dangerous leakage here: they sit on the *race* table, so a naive join propagates them silently to every runner, and is literally the saddlecloth number of the winner. They are dropped at load time, before any merge.

and are the *final* tote odds. They are not target leakage in the strict sense — the price is fixed at post time — but they are excluded from the feature set for three reasons. The closing price aggregates every bettor's information, including late money, so a model built on it largely re-learns the crowd. The resulting system would be useless in deployment, where a prediction is wanted before the market settles. And mixing the market price with a fundamental signal makes our research question unanswerable, since it asks whether pre-race attributes alone carry information. Following the two-stage practice of the racetrack efficiency literature [?], the odds are retained only as an *independent market baseline.*

TABLE I: Comparison of Related Work

Ref.	Ano	Tecnica	Dataset / entorno	Metrica principal	Resultado	Limitacion identificada
Eje A — dominio (Int. 1)						
Eje B — tecnicas (Int. 2)						
Eje C — dataset (Int. 3)						

TABLE II: Condensed data dictionary

Fields	Meaning	Avail.
— one row per runner		
, ,	Race key, saddlecloth number, horse key	PRE
, ,	Age, country of origin, sex/type (gelding, mare, colt)	PRE
,	Official HKJC rating before the race; declared equipment	PRE
, ,	Weight of horse plus equipment, assigned handicap weight, barrier	PRE
,	Pseudonymous identifiers of the connections	PRE
,	Final (starting-price) tote odds	MKT
,	Finishing position and win indicator (the target)	POST
,	Beaten margin and total running time	POST
, ,	Sectional positions, margins and split times	POST
— one row per race		
, ,	Race key, meeting date, race number in the meeting	PRE
, ,	Course, rail configuration, turf or all-weather	PRE
,	Distance in metres, reported track condition	PRE
, ,	Class, eligible rating band, advertised prize money	PRE
,	Sectional and cumulative times	POST
,	Winning combinations and tote payouts	POST

TABLE III: Excluded fields and the reason for exclusion

Field(s)	Reason
	The target in ordinal form; derives from it
,	Measured at the finish line
, ,	Describe the race while it is run
(races)	Sectional times of the race itself
,	Encode the identity of the placed runners
,	Tote payouts, computed from the result
,	Not target leakage, but closing market prices; kept only as an external baseline

Most of the signal lives in historical form rather than in the raw row: a horse's prior win rate, jockey and trainer strike rates, days since the last start, class movement. Every such feature uses an expanding window over races strictly earlier in time. The frame is sorted by (entity,) and the aggregate taken as within the group, so a race never contributes to its own feature; no statistic is computed over the full table. Debutants make this explicit: 5.5 % of rows have no prior start, so these features are null by construction and receive a "no history" indicator rather than an imputed mean, which would assign an average record to an unraced horse.

The split is temporal *and* grouped by . A random split would place runners of the same race on both sides, letting the model exploit the one-winner-per-race constraint, and would let future meetings inform past ones. Cut points sit at the 70th and 85th percentiles of the date distribution: training through 16 February 2003 (55 658 runners), validation through 20 June 2004 (11 878), test thereafter (11 911). The script asserts that no race spans more than one split, and the count is zero. Model selection uses forward chaining, never shuffled k-fold. Imputers, encoders and scalers are fitted inside a on training folds only; where high-cardinality identifiers are target-encoded, the encoding is computed out-of-fold with prior smoothing, since a naive target encoding fitted on the whole training set is itself leakage.

C. Data Quality and Imputation

Missingness arises from four distinct mechanisms, and collapsing them into one imputation rule would be a mistake.

Structural absence. The later sectional columns are null for short races, which simply have fewer timing sections: , and are missing in 33 221 rows, rising to 78 151 for the sixth section. These are not missing values — the quantity does not exist — and imputing them would fabricate a measurement. All are post-race and are dropped regardless.

Systematic absence in a pre-race field. is null for 462 races, or 7.3 % of the file, and the gap is not scattered: every affected race falls between 20 July 1997 and 18 March 1998, a contiguous block at the start of the record. This is missingness by collection period, not at random, so mean or median imputation over the whole file would import a late-period price level into early races. Because the split is temporal, the entire block sits inside the training window, which limits but does not remove the problem. is therefore kept with an explicit "unrecorded" indicator, and imputation, where needed, is performed within rather than globally, since prize money is largely a function of class.

Sporadic absence. and are missing in two rows each and are encoded as an explicit level. Dropping those rows is specifically rejected: deleting a row removes a runner from a race, changing the size of the field and breaking the one-winner-per-group structure the model depends on. Mode imputation is rejected because it invents a country of origin the model may then use. is missing in 3 735 rows, in a way related to field size and pool availability rather

than at random; the column is excluded anyway, but mean imputation on a variable of that kind would introduce bias rather than remove it.

Sentinels disguised as data. Several columns use to encode "not available", which never appears as a null, so any mean or correlation computed before cleaning is silently corrupted. The affected fields are (89 cells) and the sectional margin and split-time columns (95 further cells). It is worth noting what is *not* a sentinel: and legitimately take the value 999, and 999 lb is an ordinary within the observed 693–1 369 lb range. A naive equality test flags all three, which is why the check is applied only where 999 falls outside the plausible range of the variable.

No univariate imputation is applied to derived historical features, because absence there carries meaning: a debutant is not an average horse. An indicator column is added instead, so a tree can split on it. Median imputation is used only for physical measurements after sentinel removal, and is fitted on training folds only.

D. Outlier Detection

Beyond the sentinels above, three anomalies were found, and only one is an outlier in the statistical sense.

Implausible physical values. Three rows carry a below 900 lb and one below 800 lb, which is outside the plausible range for a racing thoroughbred and is treated as a recording error. (103–133 lb) and (2–10 years) fall within legitimate ranges and are kept; the older horses are rare but real.

An impossible category value. One row has = 15 although no field in the dataset exceeds 14 runners. With a single occurrence this is a recording error rather than a pattern, but it is reported because a model treating as an ordinal would extrapolate from it.

Censoring. is capped at 99.0, and 9.4 % of runners sit exactly at that ceiling. This is right-censoring, not an outlier: the recorded value is an upper bound, so the ranking among longest-priced runners is arbitrary and the market baseline must say so.

The treatment policy follows from the group structure: rows are never deleted, because removing a runner alters the race it belongs to. Anomalies are neutralised at the cell level — the value becomes and a missing indicator is added — so each race keeps its full field.

E. Class Imbalance

Fields range from 4 to 14 runners with a mean of 12.5 and a median of 12, so positives account for 8.0 % of rows, bounded between 7.1 % and 25 % depending on field size. The one-winner-per-race invariant holds in 6 336 races; the remaining 12 (0.19 %) record two winners. These are dead heats, not recording errors. They are retained, since discarding them would bias the sample toward decisive finishes, and the top-1 metric counts a hit when the highest-ranked runner is any of the joint winners.

The imbalance is *structural* rather than a sampling defect: it is a property of the task, not a distortion to correct. That rules out the usual remedies. Synthetic oversampling is rejected on three grounds — a synthetic runner belongs to no race, so it destroys the group structure the evaluation depends on; it breaks the constraint that win probabilities within a race sum to one; and interpolation is meaningless where , and have no midpoint. Random undersampling fails for the same reason: removing competitors changes how hard the race was to win. Class weighting is admissible for the pointwise variant but distorts the output probabilities, so scores must be recalibrated on the validation split before any log loss is reported. We prefer not to treat the problem as imbalanced at all, but to model the race as a group and normalise scores within it, making the base rate a consequence of field size rather than a nuisance parameter.

The metric plan follows. Accuracy is meaningless: a model that never predicts a winner scores 92.0 %. Primary metrics are race-level top-1 accuracy and mean reciprocal rank; calibration is measured by log loss and Brier score; discrimination is reported as ROC-AUC within races. Two reference points anchor the scale. Selecting at random within the field yields 8.1 % top-1 accuracy, and ranking by shortest final odds yields 29.8 %, so the market is 3.7 times better than chance. A model failing to beat the first is broken; the second is the market benchmark of Section **??**, and closing the gap to it using pre-race information alone is the substantive question of this work.

F. Distributions and Correlations

Field size is concentrated at the upper end — a median of 12 against a maximum of 14 — which governs the base rate above and motivates reporting metrics per race rather than per row. The barrier effect is clear and monotone: runners drawn in stall 1 win 10.5 % of the time against 6.0 % for stall 13, a factor of 1.7 across the range, with the decline steady rather than concentrated at the extremes.

The effect is not uniform across conditions. Splitting by course, the advantage of an inside draw (stalls 1–4) is roughly twice as large at Happy Valley, where those runners win 11.0 % against 7.3 % for the rest of the field, as at Sha Tin, where the same comparison is 8.9 % against 7.2 %; this is consistent with the tighter Happy Valley circuit. By distance, however, the effect is not monotone: the gap is 2.8 points below 1 200 m and 3.1 points above 1 600 m, but only 1.3 points in between. Treating as a plain ordinal, or interacting it linearly with distance, cannot represent that shape, which is the concrete argument for a draw × distance × venue interaction with distance entered categorically.

Among the pre-race numeric features the notable structure is the near-collinearity between and , expected because handicap weight is assigned from the official rating; harmless for gradient-boosted trees, but it must be handled in the logistic-regression baseline by dropping one term or by regularisation.

TABLE IV: Diagnosis and resulting decisions

Finding	Action taken
48 columns encode post-race information	Dropped at load time, before any join
Payout fields sit on the race table	Removed before the merge
Closing odds beat chance by 3.7×	Excluded as a feature; kept as external baseline
Historical form is the main signal; 5.5% debutants	Expanding windows with ; explicit no-history indicator
Runners of one race are not independent	Temporal split grouped by ; zero races span two splits
used as a sentinel in 184 cells	Converted to , but only where 999 is outside the variable's plausible range
missing for a contiguous 1997–98 block	Indicator column; imputation within , never global
4 implausible weights, 1 impossible	Cell-level neutralisation; rows never deleted
censored at 99.0 (9.4% of runners)	Documented as a limit of the market baseline
12 dead heats, 1 race without runners	Dead heats retained with an adjusted top-1 rule; empty race dropped
8.0% positives, one winner per race	No resampling; within-race normalisation
collinear with	Regularisation in the linear baseline

G. Findings and Derived Actions

Table ?? condenses the analysis into the concrete decisions it produced, each of which is carried into the experimental protocol of Section ??.

VII. Ethical Considerations

Two features of this problem make an early ethical review necessary rather than ceremonial. The application domain is a regulated gambling market, and the underlying activity has contested animal-welfare implications. Both risks are properties of the domain and cannot be engineered away; what can be decided in advance is the scope of the artefact and the claims made about it. This section states the risks we identified before training and the specific limits adopted in response. The AI usage disclosure covers the separate question of tooling, and the Stage 3 report revisits these issues with the deployed agent in view.

A. Scope of the artefact

The system estimates a win probability for each runner from information published before a race, and ranks the field accordingly. It is a retrospective study on historical records from 1997–2005. It does not consume live odds, holds no connection to any betting operator, and includes no staking, bankroll or bet-selection component. The objective function is probabilistic accuracy, not return on investment: the model is selected and reported on log loss, Brier score, top-1 accuracy and mean reciprocal rank, and never on simulated profit. This is a methodological choice as much as an ethical one — optimising for return would make results dependent on assumptions about the tote pool that the dataset does not support — but it is also the boundary that keeps the project an exercise in probability estimation rather than a wagering tool.

B. Gambling-related harm

The most obvious risk is dual use. A calibrated win probability is directly actionable in a betting market, and Hong Kong hosts one of the world's largest per-capita betting turnovers, with problem gambling recognised as a public-health concern there. Publishing an accurate model lowers the cost of participating in that market for anyone who finds it.

We do not claim this risk is eliminated; the mitigations bound it. The repository carries an explicit non-deployment statement and no interface to any betting platform. No trained model weights are published in a form intended for reuse as a tipping service. The Stage 3 agent is scoped to explaining and comparing predictions — describing which pre-race attributes moved a given estimate, and how the model's ranking compares with the market's — and declines requests for staking advice, stake sizing or bet recommendations, with that refusal treated as a functional requirement to be tested rather than a disclaimer in a README. Where results are compared against market prices, the comparison is framed as a test of informational efficiency, following the standard treatment in that literature, and not as evidence that a profitable strategy exists.

A residual risk remains and we state it plainly: any work that improves the informational efficiency of a betting market is of some use to people betting in it. Declaring the limit does not remove it.

C. Animal welfare

Thoroughbred racing carries documented welfare concerns, including racing-related injury and the fate of horses after their careers end. This project does not model, optimise or recommend anything at the level of an individual animal: it makes no recommendation about how a horse should be trained, campaigned or run, and it produces no incentive directed at a trainer or owner. It reuses records of races that were run two decades ago. That said, the work belongs to an ecosystem whose economics rest on the racing industry, and we note the tension rather than resolve it. Nothing in the analysis should be read as a position on whether the sport ought to exist.

D. Identifiable individuals

Jockeys and trainers appear as pseudonymous integer identifiers, but the pseudonymisation is weak: because the underlying race records are public, an identifier can be relinked to a named professional by matching dates, venues and results. The features we build include jockey and trainer strike rates, so the model assigns what amounts to a performance score to real people whose livelihood depends on their reputation.

Three limits follow. Identifiers stay pseudonymous in every published output, including figures and appendices. No

ranking of individual jockeys or trainers is produced or published, even though the intermediate features would permit one. Feature-importance analysis is reported at the level of feature families — "jockey history", "barrier", "weight" — rather than for specific individuals. Horse identity raises no comparable concern, and is used without restriction.

E. Representativeness and the limits of the claim

The data come from two courses under a single regulator, within a fixed historical window, and under Hong Kong's particular handicap and class system. Results generalise to that setting and no further; presenting them as a general statement about racing would be an overclaim. There is also drift within the window itself — rule changes, track work, and turnover among jockeys and trainers — which is one reason the evaluation uses a temporal split (Sec. **??**): it makes degradation over time visible instead of averaging it away, at the cost of a harder test set. Any bias present in the historical records, including systematic differences in how horses of different origins or classes were rated and handicapped, is reproduced by a model fitted to them.

F. Calibration and overclaiming

A win probability presented without its uncertainty invites misuse, and the temptation in this domain is to read a poorly calibrated score as a statement of fact. Racing outcomes carry substantial irreducible uncertainty: the best achievable performance is far from perfect prediction, and a model reporting high confidence is more likely miscalibrated than skilled. We therefore report calibration explicitly through log loss, Brier score and reliability diagrams alongside ranking metrics, and state the irreducible-uncertainty limit in the abstract and the conclusions rather than only in an appendix.

G. Provenance and licensing

The dataset is released under a CC0 1.0 Universal public domain dedication, which places no restriction on reuse and requires no attribution; we nonetheless cite the source, since scholarly attribution is a separate obligation from the licence.

AI Usage Disclosure